



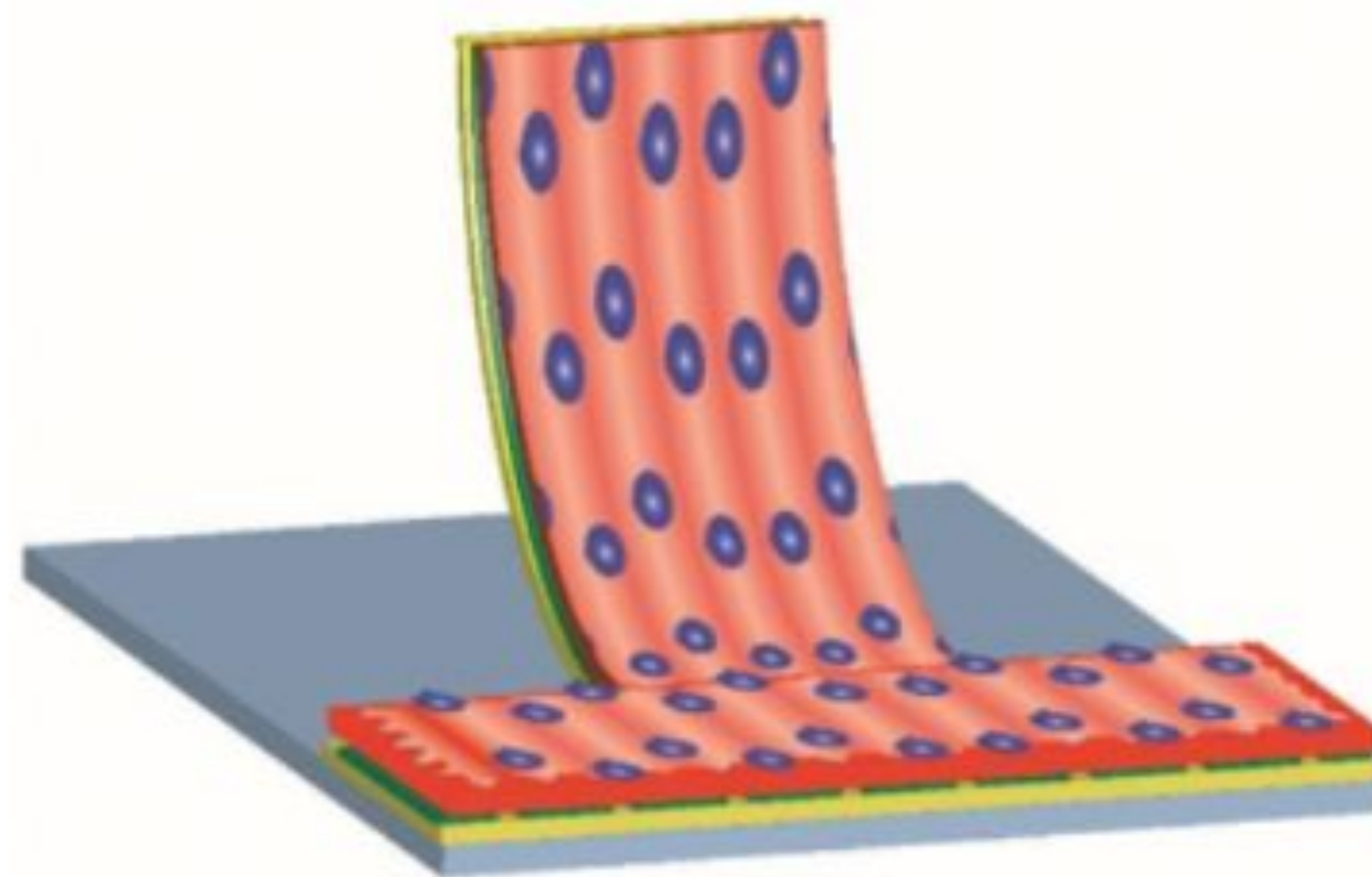
Optimization of Heart-on-a-Chip Fabrication: PDMS Spin Coating and Laser Cutting

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Introduction

- Cardiovascular diseases are the leading cause of death in the United States, but traditional research models (animal studies, patient observation) cannot isolate specific cellular mechanisms.
- Heart-on-a-chip technology effectively replicates human cardiac tissue, enabling the study of disease mechanisms in controlled environments.
- The two steps that are crucial to the fabrication of consistent and reliable hearts on a chip are pdms spin coating and laser cutting

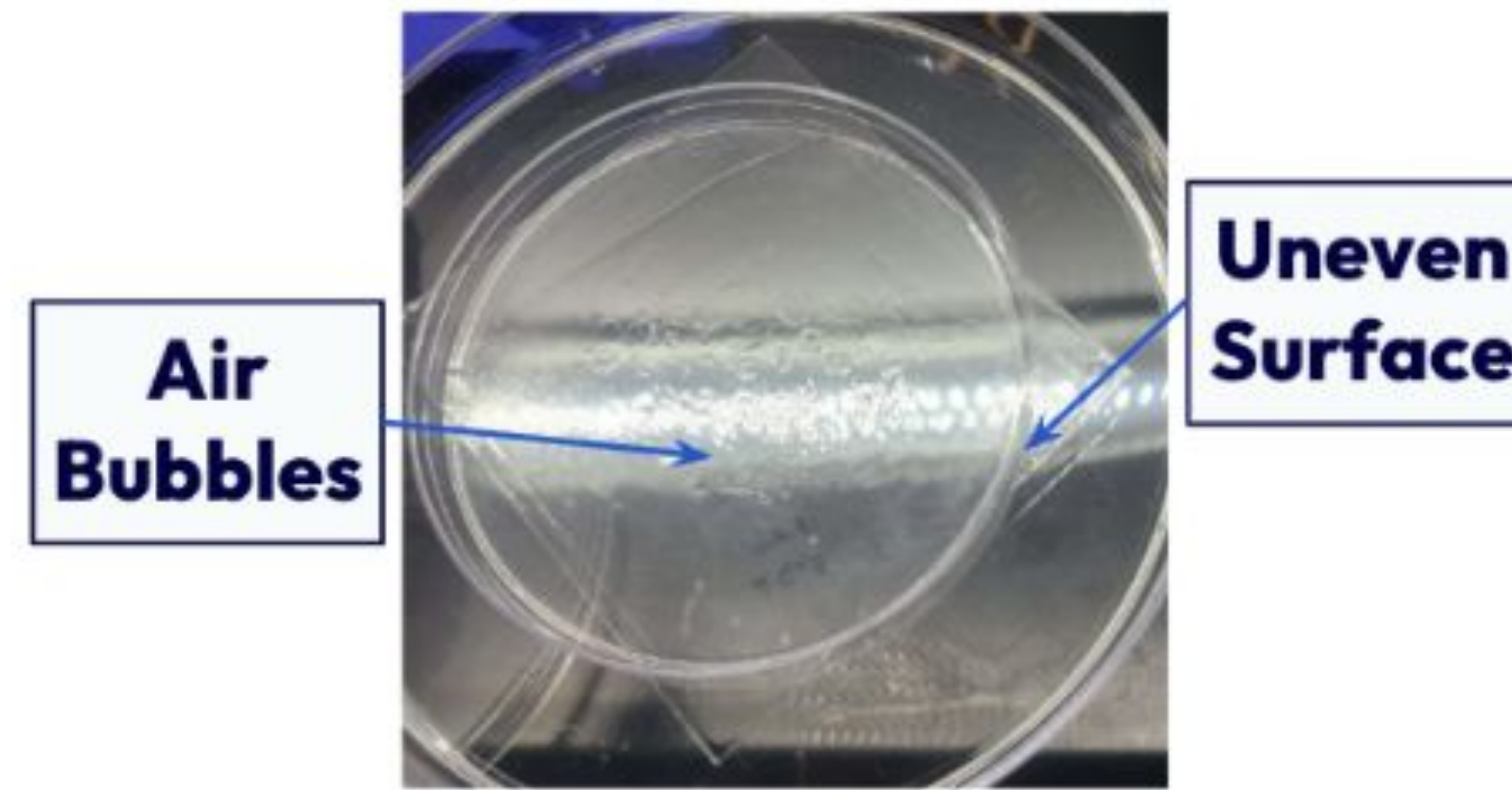


Objective

- PDMS Spin Coating:** Optimize PDMS thickness to improve cell adhesion and experimental reliability
- Laser Cutting:** Update laser cutting protocol to ensure glass coverslips are always oriented properly when scoring PDMS to ensure Pipaam lines stay clean. Optimize power settings to ensure the glass coverslips are fully cut while avoiding breakage.

Methods

PDMS Spin Coating



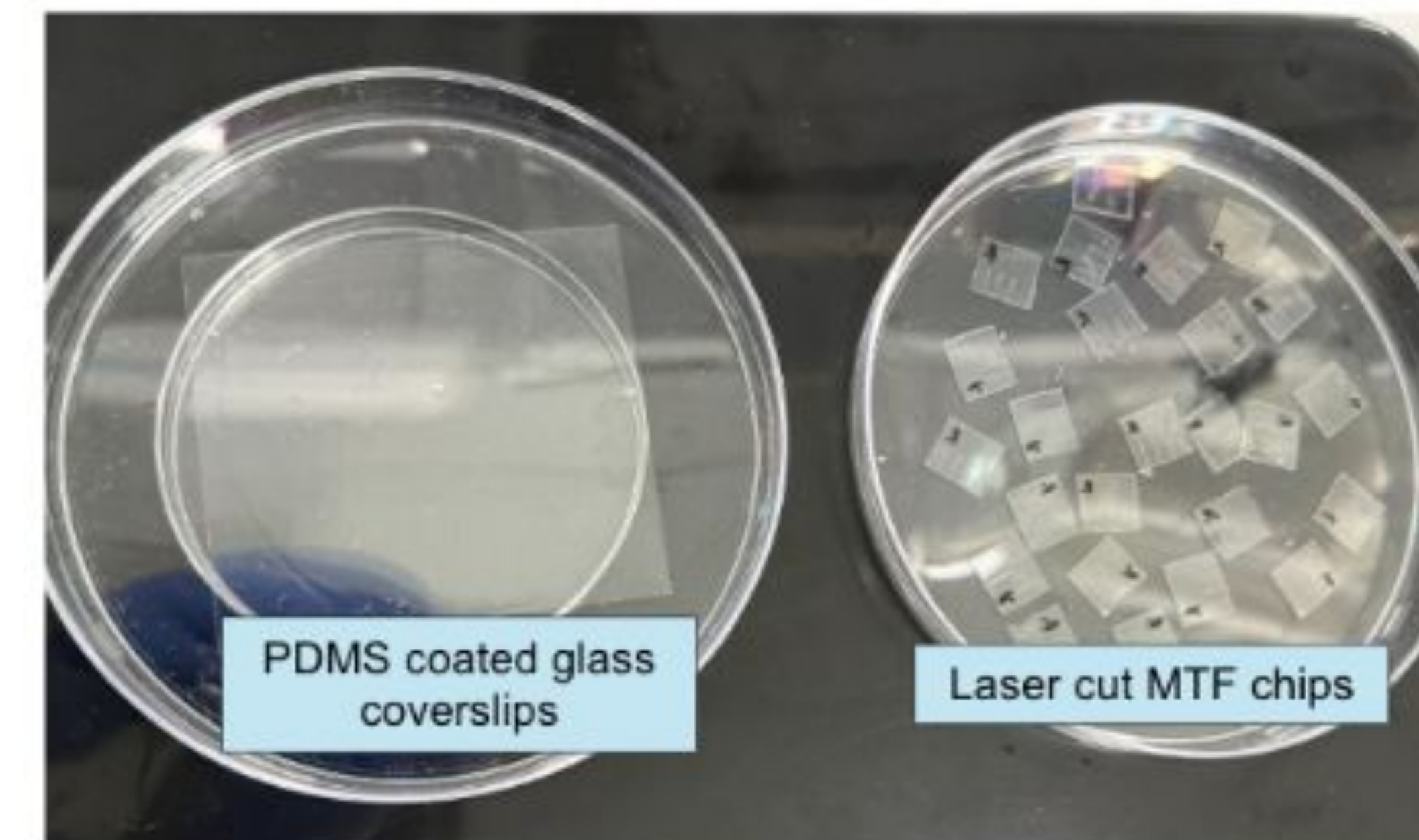
Inconsistent PDMS Thickness

#1	#2	#3
10.61	8.66	6.57
14.80	7.84	6.45
7.80	5.36	7.69

Range: ~10

- Hypothesis:** Moisture contamination in the PDMS solution
- Proposed Solution:** Place solution in oven and on vacuum desiccators for varying durations of time to draw out the moisture prior to spin coating

Laser Cutting

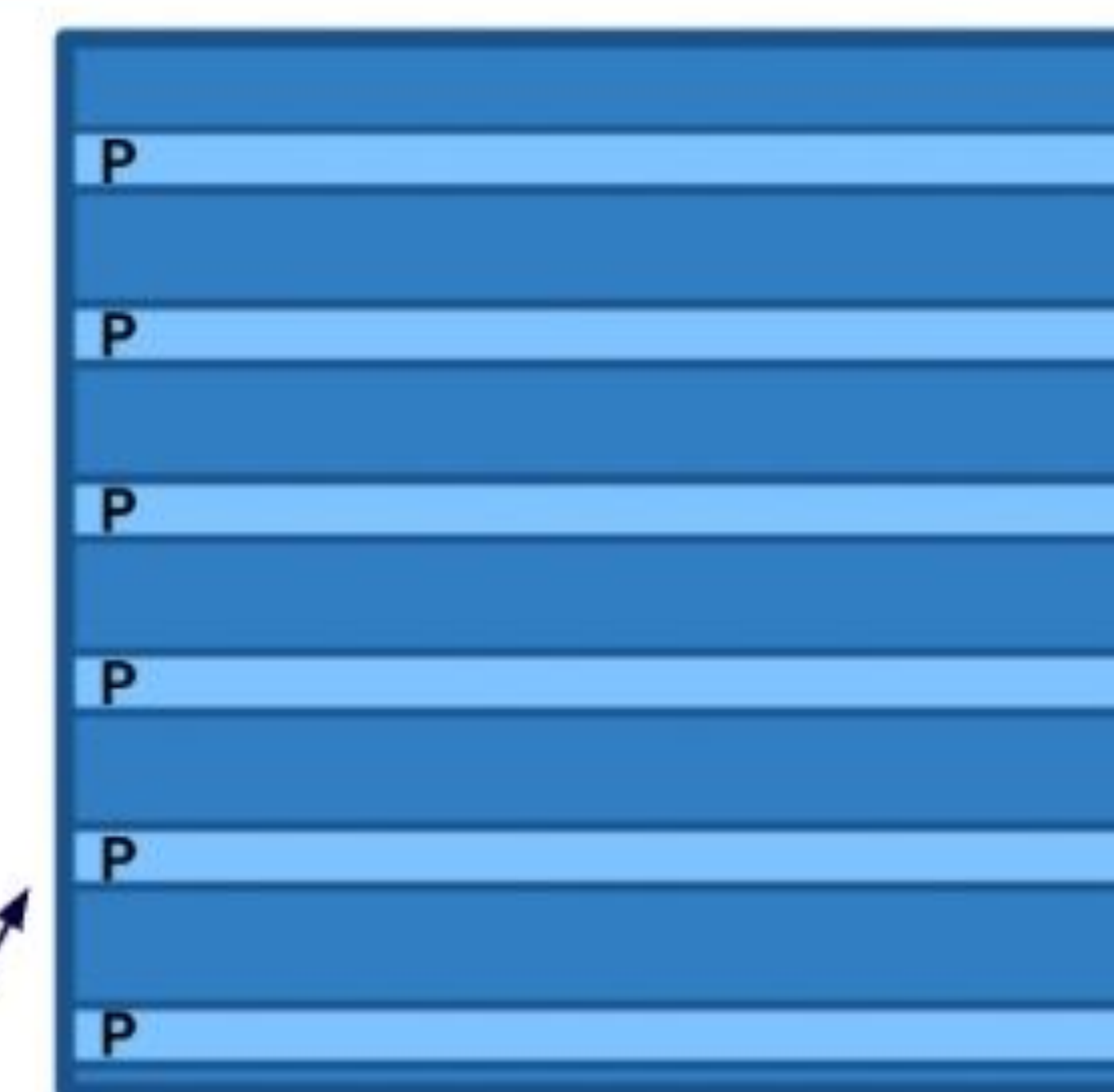
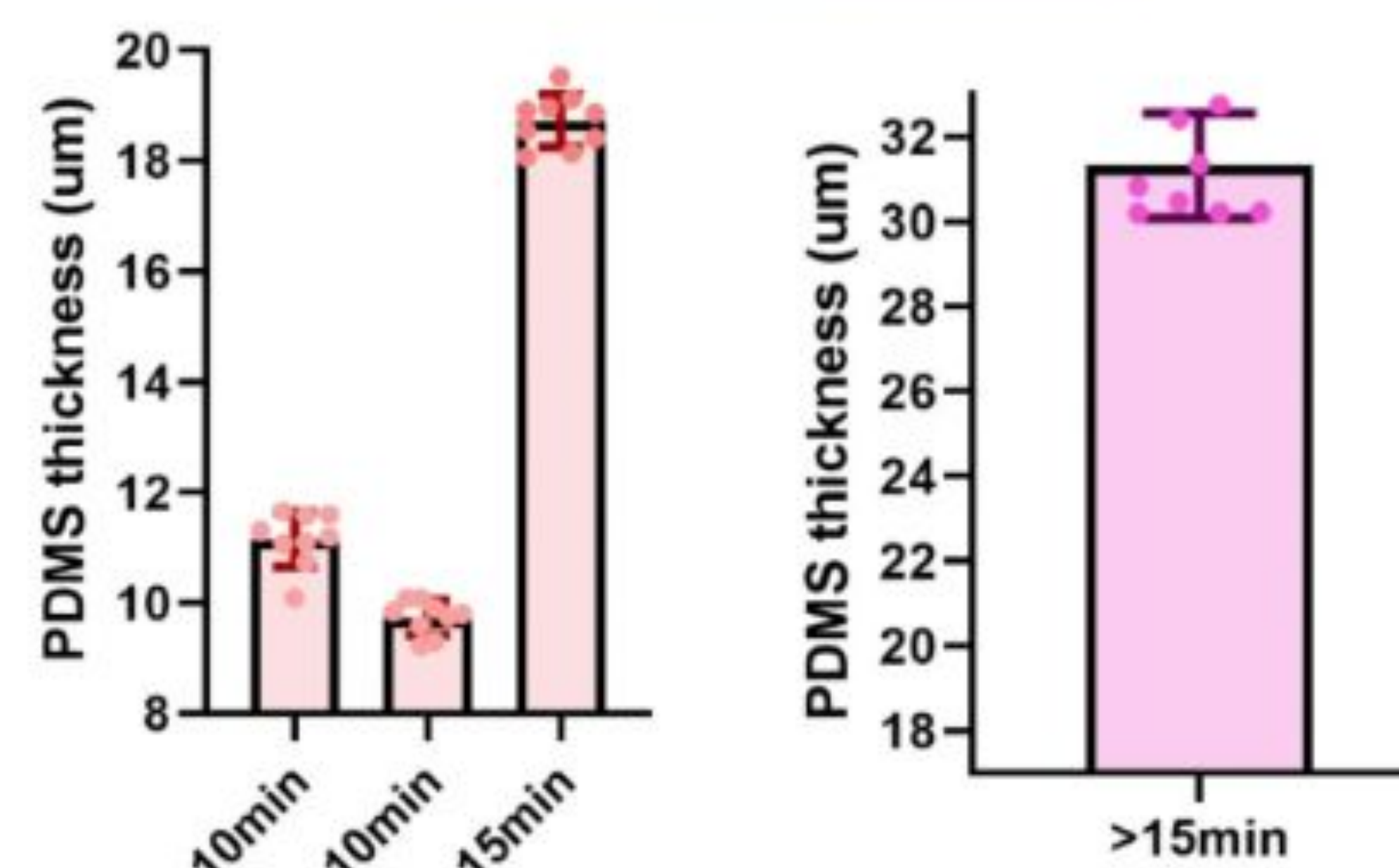
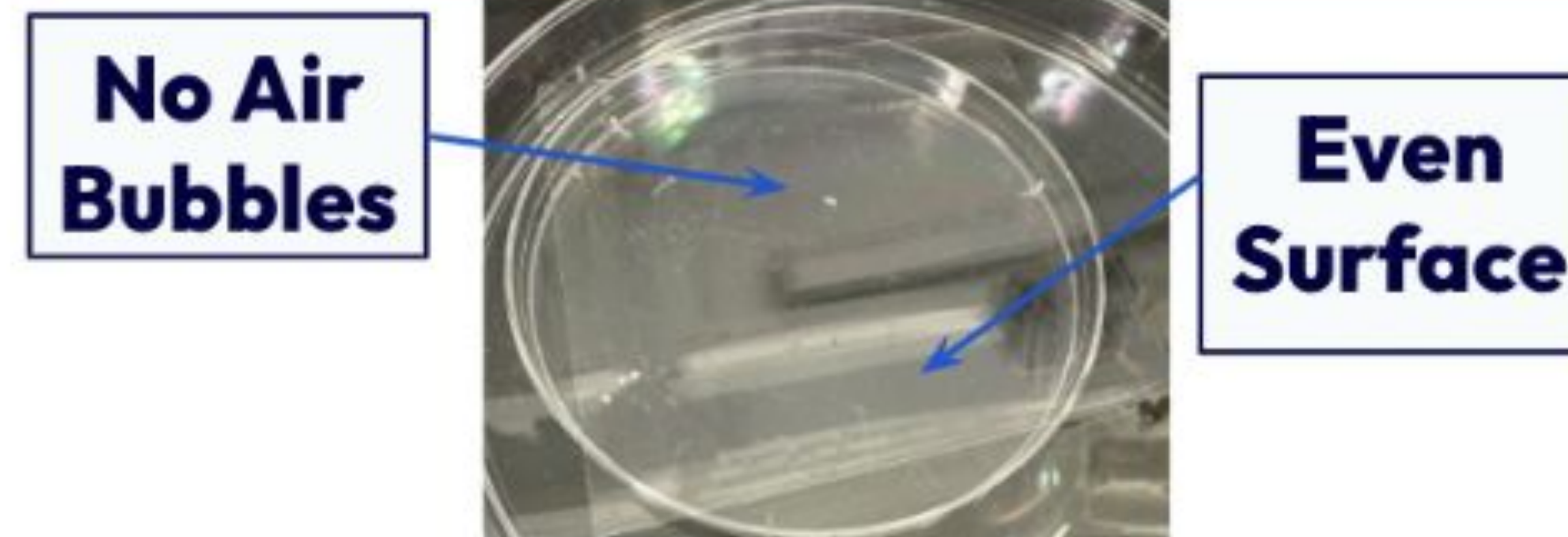


MTF lines being scored in the incorrect location

MTF films do not contract

- Hypothesis:** Current protocol leaves room for user error.
- Proposed Solution:** Update protocol to include more steps in order to minimize error throughout the process.

Results



Written P on Coverslip to Ensure Proper Orientation

Small Space Remains Below Last Pipaam Line

Future Direction

- While we have made significant progress with our project, it is far from over.
- Our optimizations have led to better results, closer to the optimal measurements and standards
- We are continuing to streamline our protocol to ensure our production of reliable heart-on-a-chip models.



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Acknowledgements

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